Vivian Nguyen

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Education

Bachelor of Software Engineering (Co-op) Sep 2022 – Apr 2028
McMaster University

Technical Skills

Languages: Python, Go, Java, TypeScript, SQL
Frameworks: React, Next.js, Spring Boot, Node.js
Databases: PostgreSQL, pgvector, MongoDB, MySQL
AI/ML: Google ADK, Gemini API, RAG, LLM Agents, Tool Calling, Vector Search
Developer Tools: Git, Linux, Docker, Jenkins, gRPC, Jest, JUnit, Playwright

Experience

Software Developer Intern May 2025 – Present
IBM *Full-time May 2025 – Aug 2026; Part-time Sep 2026 – Present*

- Developed risk management, asset inventory, and vulnerability assessment features for IBM Guardium Data Security using React, TypeScript, TanStack Router/Query, and the Carbon Design System.
- Built and maintained Go backend services with gRPC, REST APIs, and MongoDB.
- Improved quality and release reliability via automated testing (Playwright, Jest, React Testing Library, accessibility), Docker dev environments, and Jenkins CI/CD, partnering with QA, design, and product.

Software Developer Intern May 2024 – Aug 2024
Gene Solutions

- Built a responsive full-stack web application with an Angular front end using TypeScript, HTML, and CSS.
- Built and implemented REST APIs using ASP.NET and Entity Framework, for data exchange between client-side and server-side applications.

Engineering Intern May 2023 – Aug 2023
Linamar, Camtac

- Collaborated with engineering and cross-functional teams to optimize production and enhance efficiency.
- Analyzed production data, such as yield rates, cycle times, and defect rates, to identify improvement opportunities.
- Helped redesign workstations, equipment layout, and material flow to improve productivity and safety.

Projects

Ledger – LLM Agent for SEC Filing Analysis | *Python, Google ADK, Gemini, PostgreSQL, pgvector*

- Built an LLM agent that answers questions about company 10-K filings with source citations, using tool calling and vector search over 1,600+ document chunks.
- Built a pipeline that turns 30 SEC EDGAR annual reports and XBRL financial data into searchable text chunks and 177 normalized metrics.
- Resolved inconsistent filing formats (table-embedded headings, repeated page headers, missing section labels) to reach full coverage of all 30 filings.

Pulse - Observability Platform | *Go, PostgreSQL, Next.js, TypeScript, Docker*

- Built a server-monitoring platform with Go agents, a Go API, PostgreSQL, and a live Next.js dashboard.
- Load-tested with 2,000+ simulated agents; batching database writes raised throughput from 8.1K to 40K pts/sec and cut p50 latency from 2.3s to under 1ms.
- Added anomaly detection (EWMA, CUSUM, z-score) that flags unusual metrics without slowing data intake.
- Tested on 3K simulated faults; at equal false-alarm rates, EWMA caught 93.5% of drifts vs. 47% for z-score.

GemScan - Android Gem Identification App | *React Native, Python, Flask, Firebase*

- React Native mobile app that identifies gemstones using an AI-powered CNN model hosted on a Flask backend.
- Implemented image-based recognition, attribute-based search, user accounts, and secure data storage in Firebase.